S.A.V.I.O.R.

INSTRUCTION MANUAL

Dive into the caves. Bring every miner home.

A game cartridge for the Atari 2600™ Video Computer System™

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Getting started

Plug the cartridge into your console **with the power switched off**, plug a joystick controller into the **left controller jack**, then switch the console on. The title screen appears with the letters **S.A.V.I.O.R.** and your pilot hanging in the cave above them. If you own a **SaveKey** (or AtariVox), plug it into the **right controller jack** as well: the game then remembers your high score and where you stopped (see *Saving your game*).

Three versions. NTSC runs at 60 Hz on NTSC consoles and TVs. **PAL60** is the same game at 60 Hz with PAL colours, for PAL consoles and TVs that accept a 60 Hz picture. **PAL** runs at 50 Hz with PAL colours for every PAL TV: the action is about a sixth slower in real time, but the power bar still lasts the same 120 seconds.

SELECT	On the title screen, each press picks the next level. The level number is shown in the score window; once you have a high score, the window shows it for a second every few seconds. With a saved game, the title opens on the level where you stopped.
RESET	Starts the game with a score of 000000, three lives and five bombs (on the saved level of a saved game, it carries on with the lives and score you had when that level began).

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THE OBJECT OF THE GAME

Your mission

Miners are trapped deep in the mines of Minas Gerais, Brazil, and the clock is running. You are **Eorley Savior**, who founded the **S.A.V.I.O.R.** company (*Strategic Aerial Vehicle Infiltration Operations Rescue*) to rescue people trapped in deep mines. Your tools: a rocket-powered jet pack, a laser built into your helmet and a bag of bombs.

Fly from room to room, find the trapped miner and **touch** him. Deeper levels hide up to **three miners**, and the level is finished only when **every miner has been rescued**. Then the rescue bonus is counted and the next, more dangerous, cave opens.

Reach the miner before the power bar runs dry, without running into anything that bites, burns, crushes or drowns you.

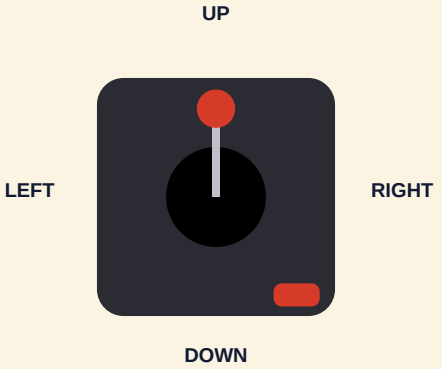
How you lose a life

- You touch a **creature**.
- You touch a **hot rock** wall, or get **crushed** by a moving wall.
- You fall into the **water** at the bottom of a flooded room.
- You are standing in the blast of your own **bomb**.
- The **power bar** runs out.

You begin with **three lives**. Lose them all and the game is **over**: you are taken back to the title screen, and the score starts again from 000000. Your **high score** is kept.

JOYSTICK AND CONSOLE

The controls



Stick UP
Fire the jet pack. Hold it to climb; let go and you float, then fall.

Stick LEFT / RIGHT
Fly sideways. The way you face is the way the laser fires.

Stick DOWN
Drop a bomb (only while you stand on the ground).

Red button
Fire the helmet laser, straight ahead.

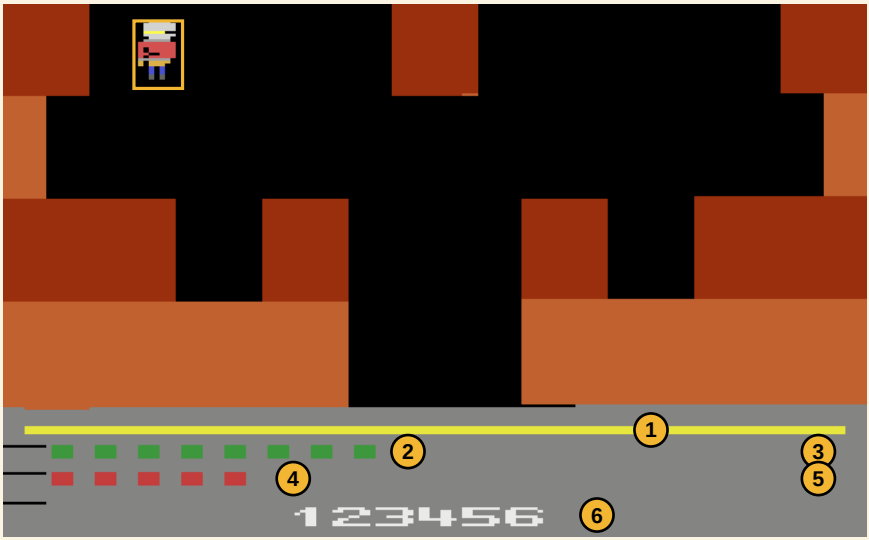
Console switches

SELECT	On the title screen, each press moves to the next level (the number shows in the score window); after the last level it wraps back to 1. **During a game**, pressing it takes you back to the title screen.
RESET	On the title screen, starts the game at the level shown: score 000000, three lives, five bombs. **During a game**, pressing it also returns to the title screen, where you can pick a level with SELECT and press RESET again.

Flying tips

- Gravity always pulls you down. **Hold UP** to burn the jet: it builds power gradually, so take-off is gentle, but once you are falling a held jet catches you quickly.
- Release UP to float for a moment, then drop. Short taps give fine control in narrow shafts.
- Move the stick **while** thrusting to slip through gaps. Walls stop you flush; they never hurt, only hot rock does.
- You can **stand** on any floor. Bombs are dropped from the ground, so land first.

Reading the screen



1	**Power bar.** Your time. It starts full and yellow; the red part grows from the right as the seconds pass. When it is all red, you lose a life. A power cell refills it.
2	**Lives.** One square per life remaining, up to eight, all green. When the last one goes, the game is over.
3	**Key.** A small yellow square at the right of the lives row. It lights up while you carry a key (not shown in this picture).
4	**Bombs.** One red square per bomb in your bag (you start with five). Drop one and a square goes out.
5	**Miners.** Purple squares: how many miners are still waiting to be rescued.
6	**Score.** Six digits, up to 999999 (it stops there). It starts again from 000000 after a game over; your best score is kept as the high score.

Your equipment

The jet pack
Powered by the stick: **UP**. Its thrust grows the longer you hold it and fades gently when you let go. It makes noise, so listen: a steady buzz means you are climbing.

The helmet laser
Hold the **red button** to send a short beam the way you face. It destroys creatures (**+50**) and switches off a lamp. It is stopped by any wall, and it cannot hurt miners, keys or power cells. Keep the button down to fire again and again.

Bombs
Press **DOWN** while standing on the floor to plant a bomb (one at a time). After about **three seconds** it explodes and the screen flashes for a second. The blast reaches a little to each side of the bomb, any height.

- A bomb **opens thin, one-cell-wide walls** (**+75**) and destroys creatures in its path (**+50**).
- It also hurts **you**. Step away from the bomb before it goes off.
- A wall that borders no other room cannot be blasted open: there is nothing on the other side.
- You carry **five bombs**. A new life does not give more, so save them for the walls that matter.

The cave

Every room is a mirror: what is on the left half is on the right. The colour of the walls changes with each level.

Solid rock
Safe to touch; it just stops you. You can stand on it.

Thin walls
A wall one cell wide. Blast it with a bomb to open the way. Stand clear.

Hot rock
Glowing red rock. **Touching it costs a life.** Keep a safe distance when you fly along it.

Locked door
A gold stripe that blocks the way. Pick up the **key** first; touching the door with the key opens it.

Phase wall
A wall that blinks on and off. It is solid when it shows and **crushes you** if it comes back while you are inside. A phase wall you have crushed through stays open for the rest of the visit.

Crusher
A wall that sweeps in from the side across a whole band every few seconds. Its **side** is deadly, so leave the band through the gap **above or below** the entrance before it closes. Its top and underside are safe: you can land on a closed crusher or bump your head against it (unless the band is hot rock).

Flooded room
The surface rises and falls a little. **Touching the water costs a life.** Cross on a raft or stay above it.

Creatures

Touching any creature costs a life. Shoot them with the laser or catch them in a bomb blast: each one is worth **50 points**. A room holds at most two of them.

Spider
Hangs at a fixed spot and bobs up and down along its thread. Time your pass for when it is at the top.

Bat
Flaps up and down on its column and never rests.

Snake
Lives inside the walls and slithers out and back along a short patrol. The laser reaches it where its head shows.

Tentacle
A water monster that bobs in a flooded room and reaches for you.

Giant moth
Patrols a room, flapping slowly but surely.

Faster and faster. Every five levels the creatures speed up: **x1.5** from level 6, **x2** from level 11, **x2.5** from level 16.

Objects

The miner
Touch him to rescue him. If other miners remain you hear a short blip and score **500**; after the last one the level ends with the rescue bonus. Check the purple squares on screen to see how many are left. Miners never move.

Key
Touch it to pick it up: a yellow square appears at the right of the lives row. It opens a locked door and stays with you when you leave the room, and when you lose a life.

Power cell
A glowing cyan cell. Touch it to **refill the power bar** completely. Each cell can be used once per level. Long levels hide a few of them.

Lamp
Touch it or shoot it and the whole room goes **dark**: only the lamp and the creatures remain visible as grey shapes. It does not cost a life, and the room stays dark until the level ends.

Raft
A log floating on the water. Land on it and it carries you across to the other side, speeding up as you ride. It moves **only while you are on it**. Step off and it waits for you.

Scoring and levels

Destroy a creature (laser or bomb)	**50**
Blow open a thin wall	**75**
Rescue a miner (when others remain)	**500**
Rescue the last miner	**Rescue bonus**

Extra lives

You earn **one extra life every 50,000 points** (at 50,000, 100,000, 150,000 and so on). You can hold at most **eight lives**; a life earned at eight is lost. The score itself stops at 999999.

The rescue bonus

When the last miner is safe, the game freezes on the rescue and counts a bonus: points for the **time left on the power bar**, a flat reward for the level, **every unused bomb** and **double points for every life** you still have. Fly quickly, save your bombs and stay alive.

Levels

There are twenty levels. Each is a network of connected rooms: leave a room through an opening on any side and you arrive in the neighbouring room. A level may have one, two or three miners (one up to level 6, two from level 7, three from level 14), doors with keys, flooded rooms, crushers and phase walls. The deeper you go, the more of these you will meet at once.

When the game ends

Lose your last life and it is **game over**: you return to the title screen and the score starts again. Clear the twentieth level and you have won; the game returns to the title too. Press **RESET** to start a new game, or **SELECT** then **RESET** to start from another level.

Saving your game

The high score

When a game ends (game over, or the twentieth level cleared), your score is compared with the **high score**; if it is higher, it becomes the new one. On the title screen the score window shows the level number and, for a second every few seconds, the high score. Nothing is shown until you have one.

The SaveKey

A **SaveKey** (or an AtariVox) in the **right controller jack** lets the game remember things after the console is switched off.

- Your **high score** is stored on it, and it is also updated every time a new level begins.
- At the start of **every level** the game saves that level, your lives and your score. You will notice a very short pause then: the game is writing to the key.
- Next time you switch on, the title screen opens on the **saved level**. Press **RESET** and you carry on with the lives and the score you had when that level began.
- If you pick another level with **SELECT**, you start that level from scratch: 000000, three lives, five bombs.
- A **game over** erases the saved game, but never the high score.
- Without a SaveKey the game plays exactly the same, it just remembers nothing.

Hints from the pros

- **Look before you leap.** Creatures follow fixed patterns; watch one full cycle before you pass.
- **Land to drop a bomb.** Bombs only come out on the ground, and you need a second or two to get clear. Plan your retreat first.
- **Use the laser, not the bombs, on creatures.** Bombs open walls; the laser costs nothing.
- **Watch the power bar.** If it is more than half red, head for a power cell or the miner. Do not explore.
- **The mirror tells you the cave.** If you can see one half of a room, you know the other half.
- **Crushers and phase walls cost a life only if you stay in.** Always leave a band by its gap, never wait inside.
- **Leave the lamp alone** unless you must: a dark room is hard to read. But remember that it stays dark.
- **A key is yours until the level ends.** Take your time to find the door; carrying it costs nothing.
- **Every miner counts.** If the purple squares are not gone, the level is not over, even when the first miner is already safe.

THANK YOU

Credits

S.A.V.I.O.R. is an Atari 2600 game written from scratch by Iuri Fiedoruk. It would not exist without the people and the work below.

The game that inspired it

H.E.R.O. (Activision, 1984, designed by John Van Ryzin) is the reference for the whole kernel: the mirrored playfield, the fixed-time scanline loop, the HUD band and the way the cartridge switches banks.

Techniques we learned from

Andrew Davie for the horizontal positioning routines. The *lo-a-rad-dragon* reference project from Rick Parrish's LORD2 repository for the fixed-point physics model. The Atari 2600 community for the 48-pixel sprite text and the player-copy tricks.

Ideas and inspiration

Felipe Zacani: the idea of multiple miners, doors and keys, and his game *Spelunk Man*.

Tools

Stella (emulator and debugger), DASM (the 6502 cross-assembler), py65 (the 6502 simulator used by the tests) and Qt (the level and model editor).

Special thanks

To my lovely wife Arismeire and daughter Katya Valeria.

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